

Math 55 quiz 2/19/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: “As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers.”

1. Let a, b, c be positive integers. Prove that $\gcd(\gcd(a, b), c) = \gcd(a, \gcd(b, c))$.

NOTE: In general, this is true no matter how you re-order the numbers. You can therefore talk about the GCD of any non-empty finite set of integers, not just pairs of them.

Solution: Let $k = \gcd(\gcd(a, b), c)$. We will prove that $k \mid \gcd(a, \gcd(b, c))$. To do this, we prove that $k \mid a$ and $k \mid \gcd(b, c)$.

- $k \mid a$: Let $q = \gcd(a, b)$. Then $k \mid q$ and $q \mid a$, so $k \mid a$. (This is a theorem that you should prove yourself!)
- $k \mid \gcd(b, c)$: By construction, $k \mid c$, and, by the same logic as in the previous bullet point, $k \mid b$. Therefore $k \mid \gcd(b, c)$.

By similar logic, $\gcd(a, \gcd(b, c)) \mid k$. Therefore the two must be equal.

2. Let x be an integer such that $x \not\equiv 0 \pmod{7}$ and $x \not\equiv 1 \pmod{7}$. Set $k = 1 + x + x^2 + x^3 + x^4 + x^5$. Show that k is divisible by 7. (*Hint: What's $(x - 1)k$?*)

Solution: We expand to get $(x - 1)k = x^6 - 1$, which, by Fermat's little theorem, is $0 \pmod{7}$ (since x is not $0 \pmod{7}$). In other words, 7 divides $(x - 1)k$. Now, 7 does not divide $x - 1$, so, by definition of primality, $\gcd(x - 1, 7) = 1$. By Euclid's lemma, therefore, $7 \mid k$.